

Uniqueness of $\mathcal{K}$ -ary lax monoidal structures on commutative relative monads

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Abstract

We show that a $\mathcal{K}$ -ary lax monoidal structure on a commutative relative monad, when one exists, is uniquely determined, provided the monad preserves the cofiltered limits that present infinite products as limits of their finite subproducts. The proof isolates two independent phenomena: a *reduction* (the structure at an infinite arity is determined by the finite structures together with the structure on the constant terminal family), which uses cofiltered-limit preservation; and the *rigidity of the terminal-family combinator* (it is forced to be an infinitary meet), which is a purely equational consequence of associativity and reindexing along surjections, valid over any cartesian base. No order-enrichment of the codomain is required.

1 Setting

Let $\mathcal{C}$ and $\mathcal{D}$ be categories with finite products, and let $J: \mathcal{C} \rightarrow \mathcal{D}$ be a functor that preserves finite products (a strong monoidal functor for the cartesian structures). Recall that a *J-relative monad* $(T, \eta, (-)^*)$ consists of a functor $T: \mathcal{C} \rightarrow \mathcal{D}$, a unit $\eta_A: JA \rightarrow TA$, and a Kleisli extension $(-)^*: \mathcal{D}(JA, TB) \rightarrow \mathcal{D}(TA, TB)$, subject to the relative monad laws of Altenkirch–Chapman–Uustalu. We say T is *commutative* when it is equipped with a commutative tensorial strength; over a base such as **Set** this strength is unique, and it induces the finite lax monoidal structures used below.

We index products by sets. Fix a class $\mathcal{K}$ of *arities* (sets) such that $\mathcal{C}$ and $\mathcal{D}$ have $\mathcal{K}$ -indexed products, J preserves them, and $\mathcal{K}$ is closed under finite products, the formation of finite arities, passage to subsets of its members, and contains the singleton 1. For an arity $K \in \mathcal{K}$ and a family $X: K \rightarrow \mathcal{C}$ we write X_k for its value and $\prod_{k \in K} X_k$ for the product (formed in $\mathcal{C}$ or in $\mathcal{D}$ as the context demands).

Definition 1. A $\mathcal{K}$ -ary lax monoidal structure on the *J-relative monad* T is a family of morphisms of $\mathcal{D}$, natural in $X: K \rightarrow \mathcal{C}$,

$$\phi_{K,X}: \prod_{k \in K} TX_k \longrightarrow T\left(\prod_{k \in K} X_k\right) \quad (K \in \mathcal{K}),$$

the inner product formed in $\mathcal{C}$ and the outer in $\mathcal{D}$, satisfying:

(U) Unitality. $\phi_{1,X} = \text{id}_{TX}$.

(A) Associativity. For $K \in \mathcal{K}$ and $(J_k)_{k \in K}$ in $\mathcal{K}$ with $M = \prod_{k \in K} J_k \in \mathcal{K}$, writing $X: M \rightarrow \mathcal{C}$ as a doubly-indexed family $X_{k,j}$ and $X'_k := \prod_{j \in J_k} X_{k,j}$:

$$\begin{array}{ccc}
 \prod_k \prod_j TX_{k,j} & \xrightarrow{\cong} & \prod_{m \in M} TX_m \\
 \downarrow \Pi_k \phi_{J_k, X_{k,\bullet}} & & \downarrow \phi_{M,X} \\
 \prod_k TX'_k & & \\
 \downarrow \phi_{K, X'} & & \\
 T(\prod_k X'_k) & \xrightarrow{\cong} & T(\prod_m X_m)
 \end{array}$$

(R) Reindexing along surjections. For every surjection $f: K \twoheadrightarrow L$ in $\mathcal{K}$ and every $X: L \rightarrow \mathcal{C}$, with the reindexing maps $f_{TX}^*: \prod_l TX_l \rightarrow \prod_k TX_{f(k)}$ and $f_X^*: \prod_l X_l \rightarrow \prod_k X_{f(k)}$ given by precomposition with f :

$$\begin{array}{ccc}
 \prod_{l \in L} TX_l & \xrightarrow{\phi_{L,X}} & T(\prod_l X_l) \\
 f_{TX}^* \downarrow & & \downarrow Tf_X^* \\
 \prod_{k \in K} TX_{f(k)} & \xrightarrow{\phi_{K,X \circ f}} & T(\prod_k X_{f(k)})
 \end{array}$$

We further require that η and the Kleisli extension be monoidal with respect to ϕ (the unit and multiplication axioms). These last two axioms play no role in what follows and we do not spell them out; the results below use only (U), (A), (R), naturality, and the finite uniqueness recorded in Hypothesis 1.

Remark 1. Reindexing is imposed only along surjections. A general reindexing $f: K \rightarrow L$ factors as a surjection (a diagonal on products, i.e. duplication of indices) followed by an injection (a projection, i.e. deletion of indices); compatibility with the injective part is equivalent to T preserving the relevant products, which is a strong and often false condition. The surjective fragment is exactly what is available in the examples of interest and is exactly what the proof uses.

We isolate the only input concerning finite arities.

Theorem 1 (Finite uniqueness; Kock, McDermott–Uustalu). *For a commutative relative monad over a suitable base (e.g. $\mathcal{C} = \mathbf{Set}$, or any base on which tensorial strengths are unique), the lax monoidal structure $\phi_{K,X}$ at every finite arity K is uniquely determined.*

We treat this as known and concentrate on the passage from finite to infinite arities.

2 The terminal-family combinator

Write $M := T1$, where 1 is the terminal object of $\mathcal{C}$. For an arity $S \in \mathcal{K}$ let $\mathbf{1}: S \rightarrow \mathcal{C}$ be the constant family at 1; since $\prod_{s \in S} 1 \cong 1$, evaluating the structure at $\mathbf{1}$ yields a morphism of $\mathcal{D}$

$$d_S := \phi_{S, \mathbf{1}} : M^S \longrightarrow M.$$

We call d_S the *combinator* at arity S . The whole of this section is a study of these morphisms; it uses only (U), (A), (R), and is valid over any cartesian $\mathcal{D}$.

Throughout we reason with generalized elements: a *point of M at stage A* is a morphism $A \rightarrow M$ in $\mathcal{D}$, and we compare points by means of the following relation.

Lemma 1. *Write $m := d_2: M \times M \rightarrow M$.*

- (i) *m is commutative, associative, and idempotent.*
- (ii) *For each object A of $\mathcal{D}$, the relation on $\mathcal{D}(A, M)$ defined by*

$$p \sqsubseteq q \quad :\Longleftrightarrow \quad m \circ \langle p, q \rangle = p$$

is a partial order, natural in A , with m computing binary meets: $m \circ \langle p, q \rangle = p \wedge q$.

- (iii) (Lower bound.) *For every $S \in \mathcal{K}$, every X , and each $s \in S$, $d_S(f) \sqsubseteq f_s$ for $f: A \rightarrow M^S$ with components $f_s := \text{pr}_s \circ f$.*
- (iv) (Constants.) *$d_S \circ \Delta_S = \text{id}_M$, where $\Delta_S: M \rightarrow M^S$ is the diagonal; equivalently d_S fixes constant families.*
- (v) (Interchange / monotonicity.) *$d_2 \circ (d_S \times d_S) = d_S \circ m^S$ up to the symmetry $M^S \times M^S \cong (M \times M)^S$. Consequently d_S is monotone: $f \sqsubseteq g$ pointwise implies $d_S(f) \sqsubseteq d_S(g)$.*

Proof. (i) Commutativity and associativity of m are the instances of (R) along the transposition $2 \rightarrow 2$ and of (A) along $3 = 1 + 1 + 1$, at the terminal family. Idempotence is (R) along the fold $\nabla: 2 \rightarrow 1$: there $f_{TX}^*: M \rightarrow M^2$ is the diagonal Δ , $f_X^*: 1 \rightarrow 1$ is the identity, and $\phi_{1,1} = \text{id}$ by (U), so the square reads $\text{id}_M = m \circ \Delta$.

(ii) Reflexivity is idempotence; antisymmetry uses commutativity ($p \sqsubseteq q$ and $q \sqsubseteq p$ give $p = m \langle p, q \rangle = m \langle q, p \rangle = q$); transitivity uses associativity. Naturality in A holds because $\sqsubseteq$ is defined by an equation between morphisms, which is stable under precomposition. That m is the meet is precisely the definition of $\sqsubseteq$ together with $m \langle p, q \rangle \sqsubseteq p, q$, which follows from (i).

(iii) Apply (A) with outer arity 2 and blocks $\{s\}$ and $S \setminus s$, so that $M = \{s\} \sqcup (S \setminus s) = S$. Since $\phi_{\{s\},1} = \text{id}$ by (U) and $\phi_{S \setminus s,1} = d_{S \setminus s}$, this gives $d_S = m \circ (\text{id} \times d_{S \setminus s})$ up to the canonical isomorphism, so $d_S(f) = m \langle f_s, d_{S \setminus s}(f|_{S \setminus s}) \rangle \sqsubseteq f_s$.

(iv) Apply (R) along $\nabla: S \rightarrow 1$ at the terminal family, exactly as in (i): $f_{TX}^* = \Delta_S$, $f_X^* = \text{id}$, $\phi_{1,1} = \text{id}$, giving $\text{id}_M = d_S \circ \Delta_S$.

(v) Both sides equal $d_{S \times 2}$ at the terminal family, computed by (A) for the two factorizations $S \times 2 = \coprod_{s \in S} 2 = \coprod_{i \in 2} S$, identified by (R) along the symmetry $S \times 2 \cong 2 \times S$. For monotonicity, suppose $f \sqsubseteq g$ pointwise, i.e. $m\langle f_s, g_s \rangle = f_s$ for all s . Then

$$m\langle d_S f, d_S g \rangle \stackrel{(v)}{=} d_S(s \mapsto m\langle f_s, g_s \rangle) = d_S(f),$$

so $d_S(f) \sqsubseteq d_S(g)$. $\square$

Lemma 2 (Rigidity of the combinator). *Let ϕ, ϕ' be two $\mathcal{K}$ -ary lax monoidal structures on T with combinators d_S, d'_S . Then $d_S = d'_S$ for every $S \in \mathcal{K}$. In other words, d_S is the S -ary meet $\bigwedge_{s \in S}$ for the order of Lemma 1, and is in particular unique.*

Proof. Both ϕ and ϕ' induce, by Lemma 1(i)–(ii), the same commutative idempotent multiplication $m = d_2 = d'_2$ on M (it is the unique finite structure at arity 2, by Theorem 1), hence the same order $\sqsubseteq$. Fix $S \in \mathcal{K}$ and a point $f: A \rightarrow M^S$. By Lemma 1(iii) for ϕ' , $d'_S(f) \sqsubseteq f_s$ for all s ; equivalently the constant family $\Delta_S \circ d'_S(f)$ satisfies $\Delta_S \circ d'_S(f) \sqsubseteq f$ pointwise. Applying d_S , which is monotone by Lemma 1(v) and fixes constants by (iv),

$$d'_S(f) = d_S(\Delta_S \circ d'_S(f)) \sqsubseteq d_S(f).$$

By symmetry $d_S(f) \sqsubseteq d'_S(f)$, so antisymmetry gives $d_S(f) = d'_S(f)$. As f was an arbitrary point (take $A = M^S$ and $f = \text{id}$), $d_S = d'_S$. $\square$

3 From finite to infinite arities

Theorem 2. *Let $J: \mathcal{C} \rightarrow \mathcal{D}$ be a finite-product-preserving functor between categories with finite products, and let T be a commutative J -relative monad satisfying Theorem 1. Let $K \in \mathcal{K}$ be an arity such that the canonical cone exhibits*

$$\prod_{k \in K} X_k = \varprojlim_{J \in \mathcal{P}_{\text{fin}}(K)} \prod_{j \in J} X_j$$

as a cofiltered limit in $\mathcal{C}$ (the transition maps being the projections), and suppose T preserves this limit. Then any two $\mathcal{K}$ -ary lax monoidal structures on T agree at K :

$$\phi_{K,X} = \phi'_{K,X} \quad \text{for every } X: K \rightarrow \mathcal{C}.$$

In particular the $\mathcal{K}$ -ary lax monoidal structure on T , if it exists, is unique.

Proof. Since T preserves the displayed cofiltered limit, the family

$$\{ T(\pi_J): T(\prod_K X) \rightarrow T(\prod_J X) \}_{J \in \mathcal{P}_{\text{fin}}(K)}$$

is a limit cone in $\mathcal{D}$, hence jointly monic. It therefore suffices to prove $T(\pi_J) \circ \phi_{K,X} = T(\pi_J) \circ \phi'_{K,X}$ for each finite $J \subseteq K$.

Fix such a J and let $X^J: K \rightarrow \mathcal{C}$ be the family with $X_k^J = X_k$ for $k \in J$ and $X_k^J = 1$ otherwise; let $h: X \rightarrow X^J$ be the family morphism that is the identity on J and the unique map $X_k \rightarrow 1$ off J . Then $\prod_K X^J \cong \prod_J X$ and $\prod_K h = \pi_J$ under this isomorphism, so naturality of ϕ in the family gives

$$T(\pi_J) \circ \phi_{K,X} = \phi_{K,X^J} \circ \prod_k T(h_k).$$

Now apply (A) to the partition $K = J \sqcup (K \setminus J)$, noting $X^J|_J = X|_J$ and $X^J|_{K \setminus J} = \mathbf{1}$:

$$\phi_{K,X^J} = \phi_{2,W} \circ (\phi_{J,X|_J} \times d_{K \setminus J}), \quad W = (\prod_J X, 1),$$

up to the canonical isomorphisms. In this expression $\phi_{J,X|_J}$ and $\phi_{2,W}$ are structures at *finite* arities, hence equal for ϕ and ϕ' by Theorem 1; and $d_{K \setminus J}$ is the combinator, equal for ϕ and ϕ' by Lemma 2. Therefore $T(\pi_J) \circ \phi_{K,X} = T(\pi_J) \circ \phi'_{K,X}$. Joint monicity of the cone gives $\phi_{K,X} = \phi'_{K,X}$. $\square$

4 Remarks

Remark 2 (What each hypothesis does). *The proof factors into two independent parts, and the hypotheses split accordingly.*

- Commutativity (Theorem 1) supplies uniqueness at finite arities; it is the only place the monad's strength enters.
- Preservation of the cofiltered limit $\prod_K = \varprojlim_J \prod_J$ makes $\{T(\pi_J)\}$ jointly monic and is what reduces the infinite arity K to its finite truncations together with the single combinator $d_{K \setminus J}$ (the proof of Theorem 2). It is genuinely used here and nowhere else.
- Reindexing along surjections (R) forces the multiplication d_2 on $T1$ to be idempotent and the combinators to fix constants (Lemma 1(i),(iv)); these are what make d_S rigid.

Remark 3 (No order-enrichment is needed). *The order $\sqsubseteq$ used in Section 2 is the intrinsic semilattice order of the idempotent commutative multiplication d_2 on $T1$; it is automatically a partial order on each hom-set $\mathcal{D}(A, T1)$, by Lemma 1(ii). At no point is $\mathcal{D}$ assumed to be order-enriched, nor is $T1$ assumed to carry an order beyond the one its own multiplication defines. The combinator is forced to be an infinitary meet purely by the equational content of (U), (A), (R). The codomain $\mathcal{D}$ need only be cartesian.*

Remark 4 (The role of $\mathbf{Set} \rightarrow \mathbf{Pos}$ and the partiality monad). *The motivating instance is $\mathcal{C} = \mathbf{Set}$, $\mathcal{D} = \mathbf{Pos}$, J the discrete-poset embedding, and T the partiality (lift) monad, for which $T1$ is the two-element semilattice and d_2 is its meet; here d_S is the S -ary conjunction $\forall$, and Theorem 2 recovers the fact that the “zip” on partial elements (defined exactly when all arguments are) is the*

unique lax monoidal structure. More generally one may replace the two-element semilattice by any dominance Σ , in which case $d_S = \forall_S$ requires $\forall_S$ to preserve Σ -openness; this is the compactness condition on the arity S , and it governs existence. Uniqueness, by contrast, needs no such condition: it is the content of the theorem that the structure is rigid wherever it exists.

Remark 5 (Generality). The base **Pos** plays no part beyond providing the relative structure: the theorem holds for any finite-product-preserving $J: \mathcal{C} \rightarrow \mathcal{D}$ between cartesian categories, with $\mathcal{C}$ carrying the cofiltered limits that present its K -fold products. The unit and multiplication axioms of Definition 1 are not used; uniqueness is a property of the underlying commutative lax monoidal functor, granted finite uniqueness. The finite-uniqueness input is the natural home for the strength-theoretic results of Kock and of McDermott–Uustalu.